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Connector including shield member and lock portion

Abstract

A connector **10** is provided with an outer conductor **20**, a plurality of dielectrics **19** to be accommodated into the outer conductor **20**, inner conductors **18** to be mounted into the respective dielectrics **19**, a shield member **25** to be arranged between the dielectrics **19** adjacent in the outer conductor **20**, and a locking portion **26** for locking the shield member **25** to the dielectric **19.**

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is based on and claims priority from Japanese Patent Application No. 2021-122074, filed on Jul. 27, 2021, with the Japan Patent Office, the disclosure of which is incorporated herein in its entirety by reference.
TECHNICAL FIELD
(2) The present disclosure relates to a connector.
BACKGROUND
(3) Japanese Patent Laid-open Publication No. 2019-003856 discloses a configuration for mounting a plurality of dielectrics assembled with center terminals into a shield shell. In the configuration of Japanese Patent Laid-open Publication No. 2019-003856, a shield plate for shielding the center terminals from each other is arranged between adjacent ones of the dielectrics. Techniques

disclosed in Japanese Patent Laid-open Publication Nos. 2008-146878 and H06-060943 are also known as techniques on connectors.

SUMMARY

(4) A connector of Japanese Patent Laid-open Publication No. 2019-003856 is configured such that the shield plate is assembled to be inserted into a gap between the dielectrics after the dielectrics are assembled with the shield shell. In assembling the shield plate in this gap, it is thought to be difficult to grip the shield plate.

(5) A connector of the present disclosure was completed on the basis of the above situation and aims to facilitate the assembling of a connector.

(6) The present disclosure is directed to a connector with an outer conductor, a plurality of dielectrics to be accommodated into the outer conductor, inner conductors to be mounted into the respective dielectrics, a shield member to be arranged between the dielectrics adjacent in the outer conductor, and a locking portion for locking the shield member to the dielectric.

(7) According to the present disclosure, a connector can be easily assembled.

(8) The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is an exploded perspective view of a connector.

(2) FIG. **2** is a perspective view of an outer conductor viewed obliquely from front.

(3) FIG. **3** is a back view of the outer conductor.

(4) FIG. **4** is an exploded perspective view showing a first dielectric and first inner conductors.

(5) FIG. **5** is an exploded perspective view showing a second dielectric, second inner conductors and a shield member.

(6) FIG. **6** is a plan view of the shield member.

(7) FIG. **7** is a perspective view showing a state where the first inner conductors are mounted in the first dielectric.

(8) FIG. **8** is a back view of the outer conductor having the first dielectrics accommodated therein.

(9) FIG. **9** is a section along A-A in FIG. **8**.

(10) FIG. **10** is a perspective view showing a state where the second inner conductors are mounted in the second dielectric.

(11) FIG. **11** is a perspective view showing a state where the shield member is mounted on the second dielectric.

(12) FIG. **12** is a section along B-B in FIG. **11**.

(13) FIG. **13** is a back view of the outer conductor having the second dielectrics accommodated therein.

(14) FIG. **14** is a section along C-C in FIG. **13**.

(15) FIG. **15** is a section along D-D in FIG. **13**.

(16) FIG. **16** is a section along E-E in FIG. **13**.

(17) FIG. **17** is a perspective view showing a state where the outer conductor is accommodated in a housing.

DETAILED DESCRIPTION

(18) In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes

may be made, without departing from the spirit or scope of the subject matter presented here.

DESCRIPTION OF EMBODIMENTS OF PRESENT DISCLOSURE

(19) First, embodiments of the present disclosure are listed and described.

(20) (1) The connector of the present disclosure is provided with an outer conductor, a plurality of dielectrics to be accommodated into the outer conductor, inner conductors to be mounted into the respective dielectrics, a shield member to be arranged between the dielectrics adjacent in the outer conductor, and a locking portion for locking the shield member to the dielectric. According to this configuration, since the locking portion locks the shield member to the dielectric, the dielectric assembled with the shield member can be accommodated into the outer conductor. Thus, it is not necessary to perform an assembling operation such as the insertion of the shield member into the outer conductor with the shield member gripped, and the connector can be easily assembled.

(21) (2) The dielectrics may include a first dielectric located on a front side in an assembling direction with the outer conductor and a second dielectric located behind the first dielectric in the assembling direction, and the shield member may be locked to the second dielectric by the locking portion. According to this configuration, if the first dielectric having the shield member locked thereto is accommodated into the outer conductor, there is a concern that the second dielectric accommodated into the outer conductor later accidentally contacts the shield member and the shield member deviates from a proper position. In contrast, if the shield member is locked to the second dielectric located behind the first dielectric in the assembling direction (i.e. if the second dielectric is accommodated into the outer conductor after the first dielectric), the shield member can be made less likely to deviate from the proper position.

(22) (3) The shield member may include a pair of sandwiching portions facing each other at a distance from each other, the second dielectric may have a pair of side surfaces facing the respective sandwiching portions, and the locking portion may be provided on the respective side surfaces of the second dielectric and the respective sandwiching portions. According to this configuration, the shield member can be easily locked to the second dielectric by being sandwiched by the sandwiching portions.

(23) (4) The locking portion may include protrusions projecting from the respective side surfaces and extending in the assembling direction and grooves formed in the respective sandwiching portions and extending in the assembling direction, and the protrusions may be fit into the grooves. According to this configuration, the shield member and the second dielectric can be prevented from being separated in a direction intersecting the assembling direction.

(24) (5) The shield member may be arranged forward of the second dielectric, and the sandwiching portions may include first restricting claws for restricting forward separation of the shield member by being locked to the second dielectric. According to this configuration, since the forward separation of the shield member from the second dielectric is restricted by the first restricting claws, the shield member can be maintained in a state locked to the second dielectric.

(25) (6) The sandwiching portions may be sandwiched by the side surfaces of the second dielectric and inner surfaces of the outer conductor with the second dielectric accommodated in the outer conductor, and the sandwiching portions may include second restricting claws to be locked to the inner surfaces of the outer conductor. According to this configuration, the second dielectric can be indirectly held in a state accommodated in the outer conductor by the sandwiching portions including the second restricting claws, and the separation of the second dielectric from the outer conductor can be suppressed.

DETAILS OF EMBODIMENT OF PRESENT DISCLOSURE

Embodiment

(26) One embodiment of the technique disclosed in this specification is described below with reference to FIGS. **1** to **17**. A connector **10** according to this embodiment is mounted on an unillustrated circuit board. In the following description, upper and lower sides shown in FIG. **1** are directly defined as upper and lower sides concerning a vertical direction. A right side and a left side

in FIG. **1** are defined as a front side and a rear side concerning a front-rear direction. A forward direction is a connecting direction of the connector **10** to an unillustrated mating connector. A back side and a front side of FIG. **1** are defined as a left side and a right side concerning a lateral direction.

(27) As shown in FIG. **1**, the connector **10** is provided with an outer conductor **20**, a housing **11** for accommodating the outer conductor **20**, a plurality of dielectrics **19** to be accommodated into the outer conductor **20**, inner conductors **18** to be respectively mounted into these dielectrics **19**, shield members **25** to be mounted on the dielectrics **19**, and locking portions **26** for locking the shield members **25** to the dielectrics **19**.

(28) [Outer Conductor]

(29) The outer conductor **20** is made of electrically conductive metal. A metal such as copper, copper alloy, aluminum or aluminum alloy is used for the outer conductor **20**. The outer conductor **20** is formed by a known method such as casting, die casting or cutting. In the case of this embodiment, the outer conductor **20** is made of die casting, specifically, made of die casting zinc or zinc alloy.

(30) As shown in FIGS. **2** and **3**, the outer conductor **20** includes four tube portions **21**, a dielectric surrounding portion **22** and a flange **23**. The four tube portions **21** are in the form of rectangular tubes with arcuate corners and extend in the front-rear direction. The dielectric surrounding portion **22** extends rearward of the rear end edges of these tube portions **21**. The flange **23** expands in directions intersecting the front-rear direction on a boundary part between the four tube portions **21** and the dielectric surrounding portion **22**.

(31) The four tube portions **21** are two pairs of the tube portions **21** arranged in the lateral direction in two upper and lower stages. Projecting portions **21**A facing and projecting toward each other are provided on respective inner surfaces of left and right side walls forming each tube portion **21** (see FIGS. **2** and **3**). The projecting portion **21**A is formed to project inward toward a front end, and the front end extends in the lateral direction (see FIGS. **2**, **9** and **14**).

(32) The plurality of dielectrics **19** are accommodated into the dielectric surrounding portion **22**. The dielectric surrounding portion **22** has an upper wall **22**A, a left wall **22**B, a right wall **22**C and a lateral center wall **22**D. The upper wall **22**A expands in a direction intersecting the vertical direction and extends rearward from the rear surface of the flange **23**. The left wall **22**B expands in a direction intersecting the lateral direction and extends rearward from the rear surface of the flange **23** while hanging down from the left end edge of the upper wall **22**A. The right wall **22**C expands in a direction intersecting the lateral direction and extends rearward from the rear surface of the flange **23** while hanging down from the right end edge of the upper wall **22**A. The lateral center wall **22**D expands in a direction intersecting the lateral direction and extends rearward from the rear surface of the flange **23** while hanging down from a laterally central part of the upper wall **22**A.

(33) As shown in FIG. **3**, the dielectric surrounding portion **22** has a left space S**1** surrounded by the upper wall **22**A, the left wall **22**B and the lateral center wall **22**D and a right space S**2** surrounded by the upper wall **22**A, the right wall **22**C and the lateral center wall **22**D.

(34) A pair of recesses **22**E are formed in each of a side surface of the left wall **22**B and a side surface of the lateral center wall **22**D facing the left space S**1**. The pair of recesses **22**E are arranged one above the other in each side surface. The pair of recesses **22**E formed in the side surface of the left wall **22**B facing the left space S**1** and the pair of recesses **22**E formed in the side surface of the lateral center wall **22**D facing the left space S**1** are recessed outward in the lateral direction (i.e. in directions away from each other).

(35) A pair of recesses **22**E are also formed in each of a side surface of the right wall **22**C and a side surface of the lateral center wall **22**D facing the right space S**2**. The pair of recesses **22**E are arranged one above the other in each side surface.

(36) As shown in FIG. **2**, two projections **24** having a cylindrical shape and projecting downward

are provided on the lower end edge of each of the left wall **22**B, the right wall **22**C and the lateral center wall **22**D. The projections **24** are positioned and inserted into mounting holes formed in the circuit board.
(37) [Housing]
(38) The housing **11** is made of insulating synthetic resin. As shown in FIG. **1**, in the form of a rectangular tube having an open front end. A back wall **11**A is provided on an end part (rear end) of the housing **11** distant from an opening end. Two holes **11**B are formed one above the other in the back wall **11**A while penetrating through the back wall **11**A in the front-rear direction. The tube portions **21** of the outer conductor **20** are inserted into the holes **11**B. With the tube portions **21** of the outer conductor **20** inserted in the holes **11**B, the flange **23** is in contact with the rear surface of the back wall **11**A from behind, whereby the outer conductor **20** and the housing **11** are coupled (see FIG. **17**).
(39) [Dielectrics]
(40) The dielectrics **19** are made of insulating synthetic resin. As shown in FIGS. **4** and **5**, the dielectrics **19** include first dielectrics **19**A and second dielectrics **19**B. The first dielectric **19**A includes a fixing portion **19**C extending in the front-rear direction and a guiding portion **19**D projecting downward from a rear side of the fixing portion **19**C. The fixing portion **19**C is formed with two through holes **19**E arranged in the lateral direction and penetrating in the front-rear direction. Recesses **19**P recessed laterally inward are formed to extend rearward from front ends in left and right side surfaces of the fixing portion **19**C (see FIG. **9**). The guiding portion **19**D is formed with an inner conductor accommodation chamber **19**F recessed forward and open rearward and downward. A partition wall **19**G extending in the vertical direction and projecting rearward is provided in a laterally central part of the inner conductor accommodation chamber **19**F to partition the inner conductor accommodation chamber **19**F into left and right sides.
(41) The second dielectric **19**B includes a fixing portion **19**H extending in the front-rear direction and a guiding portion **19**J extending downward from a rear side of the fixing portion **19**H. The fixing portion **19**H is formed with two through holes **19**K arranged in the lateral direction and penetrating in the front-rear direction. Recesses **19**R recessed laterally inward are formed to extend rearward from front ends in left and right side surfaces of the fixing portion **19**H (see FIG. **14**). The guiding portion **19**J is formed with an inner conductor accommodation chamber **19**L recessed forward and open rearward and downward. Laterally outer surfaces of the guiding portion **19**J are a pair of side surfaces **19**T. That is, the second dielectric **19**B has the pair of side surfaces **19**T. A partition wall **19**M extending in the vertical direction and projecting rearward is provided in a laterally central part of the inner conductor accommodation chamber **19**L to partition the inner conductor accommodation chamber **19**L into left and right sides. A vertical dimension of the guiding portion **19**D of the first dielectric **19**A is set shorter than that of the guiding portion **19**J of the second dielectric **19**B.
(42) Two protrusions **19**N and a recess **19**Q, which constitute the locking portion **26**, are formed on each of the side surfaces **19**T of the guiding portion **19**J. The two protrusions **19**N project laterally outward and extend in the front-rear direction. The two protrusions **19**N are arranged one above the other on the side surface **19**T. The recess **19**Q is recessed laterally inward and extends rearward from a front end (see FIG. **12**). The recess **19**Q is arranged between the two protrusions **19**N in the vertical direction.
(43) [Inner Conductors]
(44) The inner conductors **18** are formed by bending a strip-like metal plate at an intermediate position. A metal such as copper, copper alloy, aluminum or aluminum alloy is used for the inner conductors **18**. As shown in FIGS. **4** and **5**, the inner conductors **18** include first inner conductors **18**A and second inner conductors **18**B. The first inner conductor **18**A includes a straight portion **28**A extending in the front-rear direction and a bent portion **28**B bent from the rear end of the straight portion **28**A and extending downward.

(45) The second inner conductor **18**B includes a straight portion **28**C extending in the front-rear direction and a bent portion **28**D bent from the rear end of the straight portion **28**C and extending downward. A dimension in the front-rear direction of the straight portion **28**A is set shorter than that of the straight portion **28**C. A vertical dimension of the bent portion **28**B of the first inner conductor **18**A is set shorter than that of the bent portion **28**D of the second inner conductor **18**B.
(46) [Shield Members]
(47) The shield member **25** is formed by bending both left and right end parts of a metal plate in the same direction. A metal such as copper, copper alloy, aluminum or aluminum alloy is used for the shield member **25**. As shown in FIGS. **5** and **6**, the shield member **25** includes a shielding body portion **25**A and a pair of sandwiching portions **25**B. The shielding body portion **25**A is a flat plate having a rectangular outer shape and expands obliquely rearward toward a lower end while intersecting the front-rear direction (see FIG. **1**). The pair of sandwiching portions **25**B are bent rearward at a right angle to the shielding body portion **25**A from the respective left and right end edges of the shielding body portion **25**A. The pair of sandwiching portions **25**B are facing each other in parallel at a distance from each other.
(48) Each sandwiching portion **25**B includes two grooves **25**C constituting the locking portion **26**, two outer projecting portions **25**D serving as second restricting claws, and one inner projecting portion **25**E serving as a first restricting claw. The two sandwiching portions **25**B are symmetrically configured with respect to a lateral center of the shielding body portion **25**A. Accordingly, the configuration of one of the left and right sandwiching portions **25**B is described and that of the other sandwiching portion **25**B is not described.
(49) As shown in FIG. **5**, an end edge of the sandwiching portion **25**B on a side distant from the shielding body portion **25**A (i.e. rear end edge) is recessed forward to form the two grooves **25**C. These grooves **25**C are arranged one above the other on the sandwiching portion **25**B. In the sandwiching portion **25**B, each of parts divided into three in the vertical direction by the two grooves **25**C is resiliently deformable in the lateral direction. An extending direction of these grooves **25**C is inclined with respect to a plate thickness direction of the shielding body portion **25**A.
(50) As shown in FIG. **6**, two outer projecting portions **25**D project in a direction away from the adjacent sandwiching portion **25**B (i.e. laterally outward). Specifically, these outer projecting portions **25**D are formed by striking to project laterally outward toward rear ends, and the rear ends extend in the lateral direction. In one sandwiching portion **25**B, one outer projecting portion **25**D is arranged above the upper groove **25**C and the other outer projecting portion **25**D is arranged below the lower groove **25**C (see FIG. **5**).
(51) The inner projecting portion **25**E projects in a direction toward the adjacent sandwiching portion **25**B (i.e. laterally inward). Specifically, the inner projecting portion **25**E is formed by striking to project laterally inward toward a front end, and the front end extends in the lateral direction. The inner projecting portion **25**E is arranged between the upper groove **25**C and the lower groove **25**C (see FIG. **5**).
(52) [Locking Portions]
(53) As shown in FIG. **5**, the locking portions **26** are provided on the respective side surfaces **19**T of the second dielectric **19**B and the respective sandwiching portions **25**B. Specifically, the locking portions **26** are constituted by a plurality of the protrusions **19**N provided to project laterally outward from the respective side surfaces **19**T of the second dielectric **19**B and a plurality of the grooves **25**C formed in the respective sandwiching portions **25**B.
(54) [Assembling Process of Connector]
(55) Next, an example of an assembling process of the connector **10** is described. The assembling process of the connector **10** is not limited to the one described below.
(56) First, as shown in FIG. **7**, two first inner conductors **18**A are mounted into the first dielectric **19**A from behind. Specifically, the respective straight portions **28**A are press-fit into the respective

through holes **19**E from behind. Along with this, the respective bent portions **28**B are accommodated into the inner conductor accommodation chamber **19**F partitioned by the partition wall **19**G. The respective bent portions **28**B accommodated in the inner conductor accommodation chamber **19**F are held not to come out from the inner conductor accommodation chamber **19**F by ribs **19**S provided in the inner conductor accommodation chamber **19**F. In this way, two first dielectrics **19**A, in each of which two first inner conductors **18**A are mounted, are prepared.

(57) Subsequently, as shown in FIG. **8**, the first dielectrics **19**A each mounted with the first inner conductors **18**A are accommodated into the outer conductor **20**. Specifically, as shown in FIG. **9**, the fixing portions **19**C of the first dielectrics **19**A are respectively press-fit into two tube portions **21** on a lower side, out of the tube portions **21** of the outer conductor **20**, from behind. At this time, the projecting portions **21**A of the tube portion **21** are inserted into the recesses **19**P of the fixing portion **19**C from the front ends of the recesses **19**P. When the press-fitting of the fixing portion **19**C into the tube portion **21** is completed, the projecting portions **21**A reach central parts in the front-rear direction of the recesses **19**P and slightly bite into both left and right side surfaces of the fixing portion **19**C and are locked. In this way, the fixing portion **19**C of the first dielectric **19**A is restricted from coming out from the tube portion **21** of the outer conductor **20**.

(58) Subsequently, as shown in FIG. **10**, two second inner conductors **18**B are mounted into the second dielectric **19**B from behind. Specifically, the respective straight portions **28**C are press-fit into the respective through holes **19**K from behind. Along with this, the respective bent portions **28**D are accommodated into the inner conductor accommodation chamber **19**L partitioned by the partition wall **19**M. The respective bent portions **28**D accommodated in the inner conductor accommodation chamber **19**L are held not to come out from the inner conductor accommodation chamber **19**L by ribs **19**U provided in the inner conductor accommodation chamber **19**L. In this way, two second dielectrics **19**B, in each of which two second inner conductors **18**B are mounted, are prepared.

(59) Subsequently, as shown in FIG. **11**, the shield member **25** is mounted on each second dielectric **19**B from front. Specifically, the shield member **25** is so oriented that the pair of sandwiching portions **25**B project rearward from the shielding body portion **25**A. Then, the respective protrusions **19**N of the guiding portion **19**J are fit into the respective grooves **25**C of the sandwiching portions **25**B from behind. Then, the front ends of the respective protrusions **19**N are brought to front end parts of the respective grooves **25**C. At this time, the respective sandwiching portions **25**B are facing along the respective side surfaces **19**T of the guiding portion **19**J. At this time, the respective grooves **25**C of the sandwiching portions **25**B extend in an assembling direction of the first dielectric **19**A with the outer conductor **20** (i.e. front-rear direction and also merely referred to as an assembling direction below). Further, the respective protrusions **19**N are also oriented to extend in the assembling direction and fit into the respective grooves **25**C.

(60) At this time, as shown in FIG. **12**, the inner projecting portions **25**E provided on the sandwiching portions **25**B are inserted rearward into the recesses **19**Q of the guiding portion **19**J from the front ends of the recesses **19**Q. When reaching positions behind the rear ends of the recesses **19**Q, the inner projecting portions **25**E slightly bite into the side surfaces **19**T of the guiding portion **19**J of the second dielectric **19**B and are locked. At this time, the shielding body portion **25**A is arranged along the front surface of the guiding portion **19**J. In this way, the inner projecting portions **25**E restrict the forward separation of the shield member **25** from the second dielectric **19**B. In this way, the shield member **25** is locked to the second dielectric **19**B by the locking portions **26** and the inner projecting portions **25**E.

(61) Subsequently, as shown in FIG. **13**, the second dielectrics **19**B each mounted with the second inner conductors **18**B and the shield member **25** are accommodated into the outer conductor **20**. Specifically, as shown in FIG. **14**, the fixing portions **19**H of the second dielectrics **19**B are respectively press-fit into two tube portions **21** on an upper side, out of the tube portions **21** of the outer conductor **20**, from behind. At this time, the projecting portions **21**A of the tube portion **21**

are inserted rearward into the recesses **19**R of the fixing portion **19**H from the front ends of the recesses **19**R. When the press-fitting of the fixing portion **19**H into the tube portion **21** is completed, the projecting portions **21**A reach central parts in the front-rear direction of the recesses **19**R and slightly bite into both left and right side surfaces of the fixing portion **19**H and are locked. In this way, the fixing portion **19**H of the second dielectric **19**B is restricted from coming out from the tube portion **21** of the outer conductor **20**.
(62) The guiding portions **19**J of the respective second dielectrics **19**B are respectively accommodated into the left space S**1** and the right space S**2**. At this time, as shown in FIG. **15**, in the left space S**1**, the left sandwiching portion **25**B extends along the side surface of the left wall **22**B facing the left space S**1** and the right sandwiching portion **25**B extends along the side surface of the lateral center wall **22**D facing the left space S**1**. Further, in the right space S**2**, the left sandwiching portion **25**B extends along the side surface of the lateral center wall **22**D facing the right space S**2** and the right sandwiching portion **25**B extends along the side surface of the right wall **22**C facing the right space S**2**.
(63) As shown in FIGS. **15** and **16**, the respective outer projecting portions **25**D of the sandwiching portions **25**B are inserted forward into the respective recesses **22**E of the dielectric surrounding portion **22** from the rear ends of the recesses **22**E. When reaching positions forward of the front ends of the respective recesses **22**E, the respective outer projecting portions **25**D slightly bite into the respective recesses **22**E and are locked. That is, the respective outer projecting portions **25**D are locked to the inner surfaces of the dielectric surrounding portion **22** of the outer conductor **20**. At this time, the inner projecting portions **25**E restrict the forward separation of the shield member **25** from the second dielectric **19**B (see FIG. **12**), and the outer projecting portions **25**D restrict the rearward separation of the shield member **25** from the dielectric surrounding portion **22** of the outer conductor **20** (see FIGS. **15**, **16**). That is, the guiding portion **19**J of the second dielectric **19**B is indirectly restricted from coming out from the dielectric surrounding portion **22** of the outer conductor **20** by the inner projecting portions **25**E and the outer projecting portions **25**D of the shield member **25**.
(64) Since the shield member **25** is mounted on the second dielectric **19**B, the shield member **25** and the second dielectric **19**B can be handled as a single component. For example, the second dielectric **19**B can be accommodated into the outer conductor **20** by mounting a lower end part of the second dielectric **19**B mounted with the shield member **25** on a surface of an unillustrated jig and moving the outer conductor **20** so that the lower end edge of the dielectric surrounding portion **22** extends along the surface of this jig. That is, in this embodiment, the shield member **25** needs not be gripped (chucked) using a special jig.
(65) Subsequently, the tube portions **21** of the outer conductor **20** are inserted into the holes **11**B of the housing **11** from behind to accommodate the outer conductor **20** into the housing **11** (see FIG. **1**). In this way, as shown in FIG. **17**, the assembling of the connector **10** is completed.
(66) As shown in FIG. **16**, with the first dielectrics **19**A and the second dielectrics **19**B accommodated in the outer conductor **20**, the shielding body portions **25**A of the shield members **25** are arranged rearward of the first dielectrics **19**A and forward of the guiding portions **19**J of the second dielectric **19**B. That is, the shielding body portions **25**A of the shield members **25** are arranged between the first dielectrics **19**A and the second dielectrics **19**B adjacent in the front-rear direction in the outer conductor **20**. The first dielectrics **19**A are located on a front side in the assembling direction with the outer conductor **20**, and the guiding portions **19**J of the second dielectrics **19**B are located behind the first dielectrics **19**A in the assembling direction.
(67) Further, as shown in FIGS. **15** and **16**, the respective sandwiching portions **25**B are sandwiched by the side surfaces **19**T of the second dielectrics **19**B and the inner surfaces of the dielectric surrounding portion **22** of the outer conductor **20** with the second dielectrics **19**B accommodated in the outer conductor **20**.
(68) Further, as shown in FIGS. **9** and **14**, with the first dielectrics **19**A and the second dielectrics

19B accommodated in the outer conductor **20**, the straight portions **28**A, **28**B projecting further forward than the fixing portions **19**C, **19**H are accommodated in the tube portions **21**. Further, as shown in FIGS. **8** and **13**, with the first dielectrics **19**A and the second dielectrics **19**B accommodated in the outer conductor **20**, the lower end parts of the bent portions **28**B, **28**D project further downward than the dielectric surrounding portion **22**. The lower end parts of the bent portions **28**B, **28**D projecting further downward than the dielectric surrounding portion **22** are mounted into the mounting holes formed in the circuit board and soldered to conductive paths for signals formed on the circuit board, thereby being electrically connected to the conductive paths for signals.
(69) Next, functions and effects of this embodiment are described.
(70) The connector **10** of the present disclosure includes the outer conductor **20**, the plurality of dielectrics **19**, the inner conductors **18**, the shield members **25** and the locking portions **26**. The plurality of dielectrics **19** are accommodated into the outer conductor **20**. The inner conductors **18** are mounted into each dielectric **19**. The shield members **25** are arranged between the dielectrics **19** adjacent in the outer conductor **20**. The locking portions **26** lock the shield members **25** to the dielectrics **19**. According to this configuration, since the locking portions **26** lock the shield members **25** to the dielectrics **19**, the dielectrics **19** mounted with the shield members **25** can be accommodated into the outer conductor **20**. Thus, it is not necessary to perform an assembling operation such as the insertion of the shield member **25** into the outer conductor **20** with the shield member **25** gripped and the connector **10** can be easily assembled.
(71) The dielectrics **19** of the connector **10** of the present disclosure include the first dielectrics **19**A located on the front side in the assembling direction with the outer conductor **20** and the second dielectrics **19**B located behind the first dielectrics **19**A in the assembling direction. The shield members **25** are locked to the second dielectrics **19**B by the locking portions **26**. According to this configuration, if the first dielectrics **19**A having the shield members **25** locked thereto are first accommodated into the outer conductor **20**, there is a concern that the second dielectrics **19**B accommodated into the outer conductor **20** later accidentally contact the shield members **25** and the shield members **25** deviate from proper positions. In contrast, if the shield members **25** are locked to the second dielectrics **19**B located behind the first dielectrics **19**A in the assembling direction (i.e. if the second dielectrics **19**B are accommodated into the outer conductor **20** after the first dielectrics **19**A), the shield members **25** can be made less likely to deviate from the proper positions.
(72) The shield member **25** of the connector **10** of the present disclosure includes the pair of sandwiching portions **25**B facing each other at a distance from each other. The second dielectric **19**B has the pair of side surfaces **19**T facing the respective sandwiching portions **25**B. The locking portions **26** are provided on the respective side surfaces **19**T of the second dielectric **19**B and the respective sandwiching portions **25**B. According to this configuration, the shield member **25** can be easily locked to the second dielectric **19**B by being sandwiched by the sandwiching portions **25**B.
(73) The locking portion **26** of the connector **10** of the present disclosure includes the protrusions **19**N projecting from the side surface **19**T and extending in the assembling direction and the grooves **25**C formed in the sandwiching portion **25**B and extending in the assembling direction, and the protrusions **19**N are fit into the grooves **25**C. According to this configuration, the shield member **25** and the second dielectric **19**B are prevented from being separated in a direction intersecting the assembling direction.
(74) The shield member **25** of the connector **10** of the present disclosure is arranged forward of the second dielectric **19**B and the sandwiching portions **25**B include the inner projecting portions **25**E for restricting the forward separation of the shield member **25** by being locked to the second dielectric **19**B. According to this configuration, since the forward separation of the shield member **25** from the second dielectric **19**B is restricted by the inner projecting portions **25**E, the shield member **25** can be maintained in a state locked to the second dielectric **19**B.

(75) With the second dielectric **19**B accommodated in the outer conductor **20**, the sandwiching portions **25**B of the connector **10** of the present disclosure are sandwiched by the side surfaces **19**T of the second dielectric **19**B and the inner surfaces of the outer conductor **20**, and the sandwiching portions **25**B include the outer projecting portions **25**D to be locked to the inner surfaces of the outer conductor **20**. According to this configuration, the second dielectric **19**B can be indirectly held in a state accommodated in the outer conductor **20** by the sandwiching portions **25**B including the outer projecting portions **25**D and the inner projecting portions **25**E, and the separation of the second dielectric **19**B from the outer conductor **20** can be suppressed.
Other Embodiments
(76) The number of the first dielectrics and the number of the second dielectrics are not limited to the numbers disclosed in the above embodiment. Further, the number of the inner conductors for each of the first and second dielectrics is also not limited to the number disclosed in the above embodiment.
(77) Unlike the above embodiment, the outer and inner projecting portions may be provided only on one sandwiching portion. Further, the number of the protrusions of the guiding portion may be three or less or five or more. Further, the sandwiching portions may be provided with protrusions projecting toward the side surfaces of the guiding portion and the guiding portion may be formed with grooves into which the protrusions are fit.
(78) Unlike the above embodiment, only the protrusions and the grooves may be provided without providing the outer and inner projecting portions. Further, only the outer and inner projecting portions may be provided without providing the protrusions and the grooves.
(79) Unlike the above embodiment, shield members may be mounted on the first dielectrics from behind.
(80) From the foregoing, it will be appreciated that various exemplary embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various exemplary embodiments disclosed herein are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. A connector, comprising: an outer conductor; a plurality of dielectrics accommodated into the outer conductor; a plurality of inner conductors mounted into each of the plurality of dielectrics; a shield member arranged between the plurality of dielectrics adjacent in the outer conductor; and a locking portion configured to lock the shield member to one of the plurality of dielectrics in a direction opposite to an assembling direction of the plurality of dielectrics with respect to the outer conductor.
2. The connector of claim 1, wherein: the plurality of dielectrics includes a first dielectric located on a front side in the assembling direction with respect to the outer conductor, and a second dielectric located behind the first dielectric in the assembling direction, and the shield member is locked to the second dielectric by the locking portion.
3. The connector of claim 2, wherein: the shield member includes a pair of sandwiching portions facing each other at a distance from each other, the second dielectric has a pair of side surfaces facing the respective sandwiching portions, and the locking portion is provided on the respective side surfaces of the second dielectric and the respective sandwiching portions.
4. The connector of claim 3, wherein: the locking portion includes protrusions projecting from the respective side surfaces and extending in the assembling direction and grooves formed in the respective sandwiching portions and extending in the assembling direction, and the protrusions are fit into the grooves.
5. The connector of claim 3, wherein: the shield member is arranged forward of the second

dielectric in the assembling direction with respect to the outer conductor, and the sandwiching portions include first restricting claws configured to restrict forward separation of the shield member by being locked to the second dielectric.
6. The connector of claim 3, wherein: the sandwiching portions are sandwiched by the side surfaces of the second dielectric and inner surfaces of the outer conductor with the second dielectric accommodated in the outer conductor, and the sandwiching portions include second restricting claws to be locked to the inner surfaces of the outer conductor.
7. The connector of claim 1, wherein the plurality of dielectrics is accommodated into the outer conductor together with the shield member in the assembling direction of the plurality of dielectrics.